

Assignment 2

Introduction to Computing For the Social Sciences

WS 2025/26 — **Deadline: 30.11.2025, 23:59**

The overlapping task of this assignment is to visualize voting behaviour of countries in the Eurovision Song Contest (ESC) using a graph/network visualisation.

You still want to learn more about the Eurovision Song Contest? Check out:

<https://www.youtube.com/watch?v=KvVb7YaQ29g>

Task 1 - Scrape Eurovision Data from Wikipedia

Write your own function (**do not use a built-in function** such as `pd.read_html`) to get the voting results of the ESC 2010, 2011, 2012 and 2013.

Task 2 - Get data from API

For every artist competing in 2012, get the date their wikipedia page was created. Using these dates explore the hypothesis that the earlier an artist's wikipedia page was created - i.e. the more popular he/she already was prior to the Contest - the more points the country got.

Hint: To explore the hypothesis use OLS regression (see documentation:

https://www.statsmodels.org/dev/generated/statsmodels.regression.linear_model.OLS.html)

Retrieve Data from the XTools Wikipedia-API (<https://xtools.wmcloud.org/api>). Also think about which errors are most likely to occur when working with an API and handle them in your code. You only need to cover 2–3 errors.

Hint: You may need to do a bit of research, since we haven't discussed every possible error in the tutorial. Some of the errors introduced might not be the most relevant for API retrieval.

Task 3 - Network Visualization

Visualize the voting results for 2012 in a network plot. Use the “kamada-kawai-layout” (see documentation:https://networkx.org/documentation/stable/reference/generated/networkx.drawing.layout.kamada_kawai_layout.html) and consider appropriate mappings (for example, how to indicate contestants vs. voters only, how to visualize the total points a country received, or how to display the assigned points).

Hint: It makes sense to restrict your plot to only the three highest points (8, 10, 12) to prevent it from becoming too cluttered.

Task 4 - Network Visualisation II

Unfortunately, the network plots for single years of the Eurovision Song Contest are not too revealing...

This might change, if data from several years is aggregated. Then, detecting communities of countries within the data is possible.

In the GitHub folder, there are Eurovision vote results from 10 years (2010 - 2020).

Note that the countries in the columns assign points to the countries in the rows. Aggregate the data meaningfully and visualize it in a network plot. Write a short interpretation of your plot.